

26 Maths CT-1 solved

Set 1

CO1: Probability and Distributions

Question 1a

If events A and B are not mutually exclusive then show that

$$P(A \cap B) \geq P(A) + P(B) - 1.$$

Solution:

We know from the Addition Theorem of Probability that:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Rearranging for the intersection:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

Since the probability of any event cannot exceed 1, we know that

$$P(A \cup B) \leq 1.$$

If we substitute the maximum possible value of $P(A \cup B)$ (which is 1) into the equation, we get the minimum possible value for the intersection:

$$P(A \cap B) \geq P(A) + P(B) - 1$$

This is also known as **Bonferroni's Inequality** for two events.

Question 1b

In a bolt factory machine M_1 , M_2 and M_3 manufacture respectively 25%, 35% and 40% of the total of their output with 5%, 4%, and 2% respectively are defective bolts. One bolt is chosen at random and it is found defective.

What are the probabilities that it was manufactured in machine M_1 and M_3 ?

Solution:

We use *Bayes' Theorem*. Let D be the event that a bolt is defective.

Machine (E_i)	Probability $P(E_i)$	$P(D E_i)$ (Defective Given Machine)	Joint Probability $P(E_i) \cdot P(D E_i)$
M_1	0.25	0.05	$0.25 \times 0.05 = 0.0125$
M_2	0.35	0.04	$0.35 \times 0.04 = 0.0140$
M_3	0.40	0.02	$0.40 \times 0.02 = 0.0080$
Total $P(D)$			0.0345

1. Probability it came from M_1 :

$$P(M_1 | D) = \frac{0.0125}{0.0345} = \frac{125}{345} \approx \mathbf{0.3623}$$

2. Probability it came from M_3 :

$$P(M_3 | D) = \frac{0.0080}{0.0345} = \frac{80}{345} \approx \mathbf{0.2319}$$

Question 1c

Given the pdf $f(x) = \frac{1}{2}e^{-x/2}, x > 0$; evaluate the distribution function.

Solution:

The cumulative distribution function (CDF), $F(x)$, is defined as $\int_0^x f(t)dt$:

$$F(x) = \int_0^x \frac{1}{2}e^{-t/2}dt$$

$$F(x) = \frac{1}{2} \left[\frac{e^{-t/2}}{-1/2} \right]_0^x = -[e^{-t/2}]_0^x$$

$$F(x) = -(e^{-x/2} - e^0) = 1 - e^{-x/2}$$

Answer: $F(x) = 1 - e^{-x/2}$ for $x > 0$.

CO2: Binomial and Normal Distributions

Question 2a

Derive the expressions of mean and variance of Binomial distribution with given parameters.

Solution:

Let $X \sim B(n, p)$. The probability mass function is $P(X = x) = \binom{n}{x} p^x q^{n-x}$.

- **Mean ($E[X]$):**

$$E[X] = \sum_{x=0}^n x \cdot \frac{n!}{x!(n-x)!} p^x q^{n-x} = np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} q^{(n-1)-(x-1)} = np(p+q)^{n-1} = np$$

- **Variance ($\text{Var}(X)$):** Using $E[X(X-1)] = n(n-1)p^2$:

$$\text{Var}(X) = E[X(X-1)] + E[X] - (E[X])^2 = n(n-1)p^2 + np - (np)^2 = n^2p^2 - np^2 + np$$

Question 2b

Player A tosses 5 coins and player B tosses 8 coins. If the coins are unbiased, what is the probability of obtaining a total of 6 heads by the two players?

Solution:

Total coins $n = 5 + 8 = 13$. Since each coin is unbiased, $p = 1/2$. We need $P(X = 6)$.

$$P(X = 6) = \binom{13}{6} \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^7 = \binom{13}{6} \left(\frac{1}{2}\right)^{13}$$

$$\binom{13}{6} = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 1716$$

$$P(X = 6) = \frac{1716}{8192} \approx 0.2095$$

Question 2c

Mean height of soldiers is 68.22 inches with a variance of 10.8 sq. inches. How many soldiers in a regiment of 1000 would you expect to be over 6 ft. tall? (Given area between $x = 0$ to $x = 0.35$ is 0.1368 and between $x = 0$ to $x = 1.35$ is 0.3749.)

Solution:

$$\mu = 68.22, \sigma = \sqrt{10.8} \approx 3.286.$$

We want $X > 6 \text{ ft} = 72 \text{ inches}$.

$$Z = \frac{72 - 68.22}{3.286} = \frac{3.78}{3.286} \approx 1.15$$

(Note: There seems to be a discrepancy with the provided Z values in the question (0.35 and 1.35). Using $Z = 1.15$ usually yields an area of ~ 0.3749 from the mean).

Using the given area from 0 to 1.35 as the relevant hint for $P(0 < Z < 1.15)$ or similar:

$$P(X > 72) = 0.5 - 0.3749 = 0.1251$$

Expected Soldiers: $1000 \times 0.1251 \approx \mathbf{125}$ soldiers.

C03: Correlation and Sampling

Question 3a

Show that the correlation coefficient r_{xy} between x and y is $(\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2) / (2\sigma_x\sigma_y)$.

Solution:

We know the variance of the difference of two variables is:

$$\sigma_{x-y}^2 = \text{Var}(x - y) = \text{Var}(x) + \text{Var}(y) - 2\text{Cov}(x, y)$$

$$\sigma_{x-y}^2 = \sigma_x^2 + \sigma_y^2 - 2\text{Cov}(x, y)$$

Rearranging for $\text{Cov}(x, y)$:

$$2\text{Cov}(x, y) = \sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2$$

Since $r_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$, we have $\text{Cov}(x, y) = r_{xy} \sigma_x \sigma_y$.

Substituting this:

$$2(r_{xy} \sigma_x \sigma_y) = \sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2$$

$$r_{xy} = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x \sigma_y}$$

Question 3b

A random sample of size 25 is taken from $N(30, 16)$. Would the probability that the sample mean lies between 25 and 35 be greater than 0.99?

Solution:

$$\mu = 30, \sigma = \sqrt{16} = 4, n = 25.$$

$$\text{Standard Error } S.E. = \frac{\sigma}{\sqrt{n}} = \frac{4}{5} = 0.8.$$

We want $P(25 < \bar{X} < 35)$.

$$\text{For } \bar{X} = 25, Z = \frac{25-30}{0.8} = -6.25.$$

$$\text{For } \bar{X} = 35, Z = \frac{35-30}{0.8} = 6.25.$$

$P(-6.25 < Z < 6.25)$ is extremely close to 1 (virtually 100%), as $Z = 3$ already covers 99.7%.

Answer: Yes, the probability is greater than 0.99.

Set 2

CO1: Probability and Distributions

Question 1a

For any two events A and B, prove that

$$P(AB) \leq P(A \cap B) \leq P(A) + P(B).$$

(Note: There is a slight typo in the question image notation $P(A + B)$; it traditionally refers to the union $P(A \cup B)$).

Solution:

1. Part 1: $P(A \cap B) \leq P(A \cup B)$

By definition of set theory, the intersection of two sets is a subset of their union: $(A \cap B) \subseteq (A \cup B)$.

Therefore, by the monotonicity of probability: $P(A \cap B) \leq P(A \cup B)$.

2. Part 2: $P(A \cup B) \leq P(A) + P(B)$

The Addition Law of Probability states:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since probability is non-negative, $P(A \cap B) \geq 0$.

Subtracting a non-negative value from the sum $P(A) + P(B)$ means the result will be less than or equal to that sum.

Hence, $P(A \cup B) \leq P(A) + P(B)$.

Conclusion: $P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B)$.

Question 1b

In a test, an examinee either guesses or copies or knows the answer to a multiple choice question with four choices. The probability that he makes a guess is $1/3$ and the probability that he copies the answer is $1/6$. The probability that his answer is correct given that he copied is $1/8$. Find the probability that he knew the answer given that he correctly answered it.

Solution:

This is a classic application of **Bayes' Theorem**. Let's define the events:

- E_1 : The student knows the answer.
- E_2 : The student guesses.
- E_3 : The student copies.
- C : The student answers correctly.

1. Identify Individual Probabilities:

Event	Probability $P(E_i)$	$P(C E_i)$ (Correct given Event)
Knows (E_1)	$1 - (\frac{1}{3} + \frac{1}{6}) = \frac{1}{2}$	1 (If he knows, he is always correct)
Guesses (E_2)	$\frac{1}{3}$	$\frac{1}{4}$ (One correct choice out of four)
Copies (E_3)	$\frac{1}{6}$	$\frac{1}{8}$ (Given in the question)

2. Calculate Total Probability $P(C)$:

$$P(C) = P(E_1)P(C|E_1) + P(E_2)P(C|E_2) + P(E_3)P(C|E_3)$$

$$P(C) = (1/2 \times 1) + (1/3 \times 1/4) + (1/6 \times 1/8)$$

$$P(C) = 1/2 + 1/12 + 1/48 = (24 + 4 + 1)/48 = 29/48$$

3. Apply Bayes' Theorem:

We need $P(E_1|C)$:

$$P(E_1|C) = \frac{P(E_1)P(C|E_1)}{P(C)} = \frac{1/2}{29/48} = \frac{1}{2} \times \frac{48}{29} = \frac{24}{29}$$

Answer: The probability he knew the answer is **24/29**.

Question 1c

Is the following a pdf?

$$f(x) = 2x, 0 < x \leq 1; \quad 4 - 2x, 1 < x \leq 2; \quad 0, \text{ elsewhere.}$$

Solution:

For a function to be a Probability Density Function (PDF), it must satisfy:

1. $f(x) \geq 0$ for all x . (Satisfied here as x is positive in given ranges).
2. $\int_{-\infty}^{\infty} f(x)dx = 1$.

Integration:

$$\int_0^2 f(x)dx = \int_0^1 2x dx + \int_1^2 (4 - 2x) dx$$

$$\begin{aligned}
 &= [x^2]_0^1 + [4x - x^2]_1^2 \\
 &= (1 - 0) + [(8 - 4) - (4 - 1)] \\
 &= 1 + [4 - 3] = 1 + 1 = 2
 \end{aligned}$$

Conclusion: Since the total area under the curve is **2** (not 1), the function is not a PDF.

CO2: Poisson and Normal Distributions

Question 2a

Derive the expressions of mean and variance of Poisson distribution with given parameter.

Solution:

The Poisson PDF is $P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$

1. Mean ($E[X]$):

$$E[X] = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

Let $k = x - 1$:

$$E[X] = e^{-\lambda} \cdot \lambda \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot \lambda \cdot e^{\lambda} = \lambda$$

2. Variance ($Var(X)$):

First, find $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{x=2}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} = \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2$$

$$Var(X) = E[X^2] - (E[X])^2 = [E[X(X-1)] + E[X]] - (E[X])^2$$

$$Var(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

Conclusion: For a Poisson distribution, Mean = Variance = λ .

Question 2b

A distribution has mean 10.4 and s.d. 1.2. Find the limits within which an observation chosen at random almost certainly lies.

Solution:

In statistics, the "almost certainly" range usually refers to the $\mu \pm 3\sigma$ rule (Chebyshev's inequality or the Empirical Rule for normal distributions).

- Lower Limit $= \mu - 3\sigma = 10.4 - 3(1.2) = 10.4 - 3.6 = 6.8$
- Upper Limit $= \mu + 3\sigma = 10.4 + 3(1.2) = 10.4 + 3.6 = 14.0$

Answer: The observation almost certainly lies between 6.8 and 14.0.

Question 2c

5000 candidates appeared... pass mark is 40, distinction is 50. Average mark is 43, S.D. is 7. Find how many expect to get simply "pass" and how many get distinction. Given $\phi(0.43) = 0.6664$, $\phi(1) = 0.8413$.

Solution:

Let X be the marks, $X \sim N(43, 7^2)$.

Standard variable $Z = \frac{X-43}{7}$.

1. Probability of Distinction ($X \geq 50$):

$$Z = \frac{50 - 43}{7} = \frac{7}{7} = 1$$

$$P(X \geq 50) = P(Z \geq 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$$

$$\text{Number of Distinction} = 5000 \times 0.1587 \approx \mathbf{793.5 \approx 794}$$

2. Probability of Pass ($40 \leq X < 50$):

$$\text{For } X = 40, Z = \frac{40-43}{7} = -3/7 \approx -0.43$$

$$P(-0.43 \leq Z < 1) = \phi(1) - \phi(-0.43)$$

Since $\phi(-z) = 1 - \phi(z)$:

$$P(-0.43 \leq Z < 1) = 0.8413 - (1 - 0.6664) = 0.8413 - 0.3336 = 0.5077$$

$$\text{Number of "Simply Pass"} = 5000 \times 0.5077 = \mathbf{2538.5 \approx 2539}$$

C03: Sampling Theory

Question 3a

Show that the mean and standard error of sample mean ($\bar{x}$) from samples of size n are $E(\bar{x}) = \mu$ and $S.E.(\bar{x}) = \sigma/\sqrt{n}$.

Solution:

Let $X_1, X_2, \dots, X_n$ be independent samples from a population with mean μ and variance σ^2 .

$$\bar{x} = \frac{1}{n} \sum X_i$$

1. **Mean:** $E(\bar{x}) = E\left[\frac{1}{n} \sum X_i\right] = \frac{1}{n} \sum E(X_i) = \frac{1}{n}(n\mu) = \mu.$

2. **Standard Error:**

$$Var(\bar{x}) = Var\left[\frac{1}{n} \sum X_i\right] = \frac{1}{n^2} \sum Var(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}.$$

$$S.E.(\bar{x}) = \sqrt{Var(\bar{x})} = \frac{\sigma}{\sqrt{n}}.$$

Question 3b

Normal population has mean 0.1 and S.D. 2.1. Find the probability that the mean of a sample size 900 will be negative.

Solution:

$$\mu = 0.1, \sigma = 2.1, n = 900.$$

Sample mean distribution: $\bar{x} \sim N(\mu, \sigma/\sqrt{n})$.

$$\text{Standard Error (S.E.)} = \frac{2.1}{\sqrt{900}} = \frac{2.1}{30} = 0.07.$$

We want $P(\bar{x} < 0)$.

$$Z = \frac{0-0.1}{0.07} = -1.428 \approx -1.43.$$

$$P(Z < -1.43) = 1 - P(Z < 1.43).$$

Given: $P(|Z| > 1.43) = 0.153$. This is a two-tailed probability.

The one-tailed probability $P(Z < -1.43)$ is exactly half of this:

$$P(Z < -1.43) = 0.153/2 = \mathbf{0.0765}.$$

Set 3

CO1: Independence and Density Functions

Question 1a

If events A and B are independent then prove that A' and B' are also independent.

Solution:

Two events are independent if $P(A \cap B) = P(A)P(B)$. To show A' and B' are independent, we must prove $P(A' \cap B') = P(A')P(B')$.

By De Morgan's Law, $A' \cap B' = (A \cup B)'$. Therefore:

$$P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B)$$

Using the addition theorem:

$$P(A' \cap B') = 1 - [P(A) + P(B) - P(A \cap B)]$$

Since A and B are independent, substitute $P(A \cap B) = P(A)P(B)$:

$$P(A' \cap B') = 1 - P(A) - P(B) + P(A)P(B)$$

$$P(A' \cap B') = (1 - P(A)) - P(B)(1 - P(A))$$

$$P(A' \cap B') = (1 - P(A))(1 - P(B)) = P(A')P(B')$$

Conclusion: A' and B' are independent.

Question 1b

A bag contains 1 B and 2 W balls. A drawing consists of taking a ball and keeping it out if it is W but putting it back if it is B. Calculate the probabilities: i) 1st drawing is W, ii) 2nd drawing is W, iii) 3rd drawing is W.

Solution:

Let W_n be the event that the n -th draw is White and B_n be Black.

Initially: Total = 3 (1B, 2W).

Draw	Scenario	Probability Calculation	Result
i) 1st	Direct draw	$P(W_1) = \frac{2}{3}$	2/3
ii) 2nd	$(W_1 \cap W_2)$ or $(B_1 \cap W_2)$	$(2/3 \cdot 1/2) + (1/3 \cdot 2/3) = 1/3 + 2/9$	5/9
iii) 3rd	Exhaustive paths*	$(W_1W_2W_3) + (W_1B_2W_3) + (B_1W_2W_3) + (B_1B_2W_3)$	19/54

*Calculation for 3rd:

$(2/3 \cdot 1/2 \cdot 0) + (2/3 \cdot 1/2 \cdot 1/2) + (1/3 \cdot 2/3 \cdot 1/2) + (1/3 \cdot 1/3 \cdot 2/3) =$
 (Note: if W is kept out, third draw probability drops to zero if two W were already picked).

Question 1c

$f(x) = k/x^2$, $(1000 \leq x \leq 1500)$. Determine k and $P(X \geq 1200)$.

Solution:

- Find k : $\int_{1000}^{1500} \frac{k}{x^2} dx = 1 \implies k \left[-\frac{1}{x} \right]_{1000}^{1500} = 1$
 $k \left[\frac{1}{1000} - \frac{1}{1500} \right] = 1 \implies k \left[\frac{3-2}{3000} \right] = 1 \implies \mathbf{k = 3000}.$
- $P(X \geq 1200)$:
 $\int_{1200}^{1500} \frac{3000}{x^2} dx = 3000 \left[\frac{1}{1200} - \frac{1}{1500} \right] = 3000 \left[\frac{5-4}{6000} \right] = \mathbf{0.5}.$

CO2: Distributions

Question 2a

Approximate i) Poisson distribution from Binomial distribution and ii) Normal distribution from Poisson distribution.

i) Poisson Approximation to the Binomial Distribution

Theorem: Let X be a Binomial random variable with parameters n (number of trials) and p (probability of success). If we let $n \rightarrow \infty$ and $p \rightarrow 0$ such that the mean np remains a constant $\lambda > 0$, then the probability mass function (PMF) of X approaches the PMF of a Poisson distribution with parameter λ .

Proof:

The PMF of a Binomial distribution is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where $k = 0, 1, 2, \dots, n$.

Since we assume $np = \lambda$, we can express p as:

$$p = \frac{\lambda}{n}$$

Substitute this p into the Binomial PMF:

$$P(X = k) = \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

We can expand the combination formula and group the terms algebraically to evaluate the limits more easily:

$$P(X = k) = \frac{\lambda^k}{k!} \cdot \frac{n(n-1)(n-2)\cdots(n-k+1)}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-k}$$

Now, we analyze the limit of each part as $n \rightarrow \infty$ (keeping k and λ constant):

1. **The first term:** $\frac{\lambda^k}{k!}$ does not depend on n , so it remains constant.
2. **The fraction term:** We can distribute the n^k denominator to each factor in the numerator:

$$\lim_{n \rightarrow \infty} \frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \cdots \frac{n-k+1}{n} = \lim_{n \rightarrow \infty} 1 \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)$$

Since k is finite, as $n \rightarrow \infty$, every term in the parenthesis approaches 1. Thus, the whole product approaches 1.

3. **The exponential term:** By the standard definition of the mathematical constant e :

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$$

4. **The final term:** Because k is a fixed finite number:

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-k} = 1^{-k} = 1$$

Multiplying these limits together yields:

$$\lim_{n \rightarrow \infty} P(X = k) = \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda} \cdot 1 = \frac{\lambda^k e^{-\lambda}}{k!}$$

This is exactly the PMF of a Poisson distribution with parameter λ .

ii) Normal Approximation to the Poisson Distribution

Theorem: Let X be a Poisson random variable with parameter λ . As $\lambda \rightarrow \infty$, the standardized variable $Z = \frac{X - \lambda}{\sqrt{\lambda}}$ converges in distribution to the Standard Normal distribution $N(0, 1)$.

Proof:

The most rigorous way to prove this is by using Moment Generating Functions (MGFs) and Taylor series expansions.

The MGF of a Poisson distribution $X \sim \text{Poisson}(\lambda)$ is:

$$M_X(t) = E[e^{tX}] = e^{\lambda(e^t - 1)}$$

We want to find the MGF of the standardized variable Z :

$$Z = \frac{X - \lambda}{\sqrt{\lambda}}$$

Using the linear transformation properties of MGFs, where

$$M_{aX+b}(t) = e^{bt} M_X(at):$$

$$M_Z(t) = M_{\frac{1}{\sqrt{\lambda}}X - \sqrt{\lambda}}(t) = e^{-t\sqrt{\lambda}} \cdot M_X\left(\frac{t}{\sqrt{\lambda}}\right)$$

Substitute the Poisson MGF into this equation:

$$M_Z(t) = \exp(-t\sqrt{\lambda}) \cdot \exp\left[\lambda\left(e^{t/\sqrt{\lambda}} - 1\right)\right]$$

$$M_Z(t) = \exp\left[-t\sqrt{\lambda} + \lambda\left(e^{t/\sqrt{\lambda}} - 1\right)\right]$$

To evaluate the limit of the exponent as $\lambda \rightarrow \infty$, we use the Taylor series expansion for e^u around 0, where $u = \frac{t}{\sqrt{\lambda}}$:

$$e^u = 1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \dots$$

Substitute $u = \frac{t}{\sqrt{\lambda}}$:

$$e^{t/\sqrt{\lambda}} = 1 + \frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + \frac{t^3}{6\lambda^{3/2}} + \dots$$

Now, substitute this expansion back into the exponent of our MGF expression:

$$\text{Exponent} = -t\sqrt{\lambda} + \lambda\left(\left[1 + \frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + \frac{t^3}{6\lambda^{3/2}} + \dots\right] - 1\right)$$

$$\text{Exponent} = -t\sqrt{\lambda} + \lambda\left(\frac{t}{\sqrt{\lambda}} + \frac{t^2}{2\lambda} + \frac{t^3}{6\lambda^{3/2}} + \dots\right)$$

Distribute the λ :

$$\text{Exponent} = -t\sqrt{\lambda} + t\sqrt{\lambda} + \frac{t^2}{2} + \frac{t^3}{6\sqrt{\lambda}} + \dots$$

The $-t\sqrt{\lambda}$ and $+t\sqrt{\lambda}$ terms cancel out perfectly:

$$\text{Exponent} = \frac{t^2}{2} + \frac{t^3}{6\sqrt{\lambda}} + \dots$$

Now, we take the limit as $\lambda \rightarrow \infty$. All terms with λ in the denominator will approach 0:

$$\lim_{\lambda \rightarrow \infty} \left(\frac{t^2}{2} + \frac{t^3}{6\sqrt{\lambda}} + \dots\right) = \frac{t^2}{2}$$

Therefore, the limit of the MGF of Z is:

$$\lim_{\lambda \rightarrow \infty} M_Z(t) = e^{t^2/2}$$

The expression $e^{t^2/2}$ is precisely the Moment Generating Function of the standard normal distribution $N(0, 1)$. By Lévy's Continuity Theorem, since the MGF of Z converges pointwise to the MGF of a standard normal distribution, the distribution of Z converges to the standard normal distribution.

Question 2b

If 3% of bolts are defective, find probability that in a sample of 200, 5 are defective. ($e^{-6} = 0.00248$).

Solution:

Using Poisson approximation ($n = 200, p = 0.03 \implies \lambda = np = 6$).

$$P(X = 5) = \frac{e^{-6} \cdot 6^5}{5!} = \frac{0.00248 \cdot 7776}{120} \approx \mathbf{0.1607}$$

Question 2c

A die is tossed 1200 times. Find probability that "sixes" lies between 190 and 210.

Solution:

$$n = 1200, p = 1/6, q = 5/6.$$

$$\mu = np = 200, \sigma = \sqrt{npq} = \sqrt{200 \cdot 5/6} = \sqrt{166.67} \approx 12.91.$$

$$\text{For } X = 190, Z = \frac{190 - 200}{12.91} \approx -0.78.$$

$$\text{For } X = 210, Z = \frac{210 - 200}{12.91} \approx 0.78.$$

$$P(-0.78 < Z < 0.78) = 2 \times 0.2823 = \mathbf{0.5646}.$$

CO3: Statistics and Sampling

Question 3a

Prove that $\sum (x_i - A)^2$ is least when $A = \bar{x}$.

Solution:

Let $S = \sum (x_i - A)^2$. To minimize, $\frac{dS}{dA} = 0$.

$$\frac{dS}{dA} = \sum 2(x_i - A)(-1) = -2[\sum x_i - nA] = 0$$

$$\sum x_i = nA \implies A = \frac{\sum x_i}{n} = \bar{x}$$

The second derivative $\frac{d^2S}{dA^2} = 2n > 0$, confirming a minimum.

Question 3b

$x \sim N(68, 2.5^2)$. Find sample size n such that $P(|\bar{x} - \mu| \leq 1) = 0.95$.

Solution:

Given $P(Z \leq 1.96) = 0.975$ (so the middle 95% corresponds to $Z = 1.96$).

We know $Z = \frac{|\bar{x} - \mu|}{\sigma/\sqrt{n}}$.

$$1.96 = \frac{1}{2.5/\sqrt{n}} \implies \sqrt{n} = \frac{1.96 \cdot 2.5}{1} = 4.9$$

$$n = (4.9)^2 \approx 24.01$$

Answer: Sample size n should be 25.